

# The Alon-Tarsi basis conjecture

OPG-152

L1 - TRACTABLE

OPEN

Combinatorics

## ORIGINAL PROBLEM STATEMENT

Conjecture If  $B_1, B_2, \dots, B_p$  are invertible  $n \times n$  matrices with entries in  $\mathbb{Z}_p$  for a prime  $p$ , then there is a  $n \times (p-1)n$  submatrix  $A$  of  $[B_1 B_2 \dots B_p]$  so that  $A$  is an AT-base.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>